6.20.33 Regular Language: GATE IT 2006 | Question: 80

Let L be a regular language. Consider the constructions on L below:

I. repeat $(L) = \{ww \mid w \in L\}$
II. prefix $(L) = \{u \mid \exists v : uv \in L\}$
III. suffix $(L) = \{v \mid \exists u : uv \in L\}$
IV. half $(L) = \{u \mid \exists v : |v| = |u| \text{ and } uv \in L\}$

Which of the constructions could lead to a non-regular language?

A. Both I and IV B. Only I C. Only IV D. Both II and III

gateit-2006 theory-of-computation normal regular-language

Answer key

6.20.34 Regular Language: GATE IT 2006 | Question: 81

Let L be a regular language. Consider the constructions on L below:

I. $\text{repeat}(L) = \{ww \mid w \in L\}$
II. $\text{prefix}(L) = \{u \mid \exists v : uv \in L\}$
III. $\text{suffix}(L) = \{v \mid \exists u : uv \in L\}$
IV. $\text{half}(L) = \{u \mid \exists v : |v| = |u| \text{ and } uv \in L\}$

Which of the constructions could lead to a non-regular language?

a. Both I and IV
b. Only I
c. Only IV
d. Both II and III

Which choice of L is best suited to support your answer above?

A. $(a+b)^*$
B. $\{\epsilon, a, ab, bab\}$
C. $(ab)^*$
D. $\{a^nb^n \mid n \geq 0\}$

gateit-2006 theory-of-computation normal regular-language

Answer key

6.20.35 Regular Language: GATE IT 2008 | Question: 35

Which of the following languages is (are) non-regular?

- $L_1 = \{0^m1^n \mid 0 \leq m \leq n \leq 10000\}$
- $L_2 = \{w \mid w$ reads the same forward and backward$\}$
- $L_3 = \{w \in \{0,1\}^* \mid w$ contains an even number of 0's and an even number of 1's$\}$

A. L_2 and L_3 only B. L_1 and L_2 only C. L_3 only D. L_2 only

gateit-2008 theory-of-computation normal regular-language

Answer key

6.21 Turing Machine (1)

✍ **Practice Test:** Test 1 (13Q)

6.21.1 Turing Machine: GATE CSE 2026 | Set 2 | Question: 3

Which one of the following statements is equivalent to the following assertion?
Turing machine M decides the language $L \subseteq \{0,1\}^*$

A. Turing machine M halts on all input strings in $\{0,1\}^*$
B. Turing machine M accepts all input strings in L
C. Turing machine M rejects all input strings in $\{0,1\}^* - L$
D. Turing machine M accepts all input strings in L and rejects all input strings in $\{0,1\}^* - L$

gatecse-2026-set2 theory-of-computation turing-machine one-mark

Answer key☟

Answer Keys

6.1.1	A;C	6.1.2	A;B	6.1.3	A	6.1.4	C	6.1.5	D
6.1.6	B	6.1.7	B	6.1.8	B;C	6.1.9	C	6.1.10	D
6.2.1	A	6.2.2	A;B;C	6.3.1	False	6.3.2	A;D	6.3.3	A
6.3.4	B	6.3.5	B	6.3.6	B	6.3.7	N/A	6.3.8	N/A
6.3.9	B	6.3.10	C	6.3.11	B	6.3.12	D	6.3.13	B
6.3.14	B	6.3.15	D	6.3.16	D	6.3.17	B	6.3.18	B
6.3.19	D	6.3.20	A	6.3.21	C	6.3.22	C	6.3.23	C
6.3.24	B;C;D	6.3.25	B;C;D	6.3.26	B	6.3.27	C	6.3.28	B;D
6.3.29	A;C;D	6.3.30	B	6.3.31	B	6.3.32	A	6.3.33	C
6.3.34	D	6.3.35	B	6.4.1	B	6.4.2	C	6.4.3	B
6.5.1	True	6.5.2	True	6.5.3	N/A	6.5.4	B;D	6.5.5	B;C
6.5.6	N/A	6.5.7	D	6.5.8	B	6.5.9	N/A	6.5.10	A
6.5.11	B	6.5.12	N/A	6.5.13	N/A	6.5.14	C	6.5.15	A
6.5.16	B	6.5.17	B	6.5.18	D	6.5.19	D	6.5.20	A
6.5.21	D	6.5.22	C	6.5.23	C	6.5.24	A	6.5.25	D
6.5.26	D	6.5.27	A	6.5.28	C	6.5.29	A;B;C	6.5.30	D
6.6.1	A	6.7.1	N/A	6.7.2	N/A	6.7.3	N/A	6.7.4	N/A
6.7.5	N/A	6.7.6	A	6.7.7	N/A	6.7.8	A	6.7.9	N/A
6.7.10	C	6.7.11	B	6.7.12	A	6.7.13	X	6.7.14	B
6.7.15	C	6.7.16	A	6.7.17	C	6.7.18	C	6.7.19	B
6.7.20	D	6.7.21	D	6.7.22	A	6.7.23	B	6.7.24	C
6.7.25	D	6.7.26	256 : 256	6.7.27	B	6.7.28	B;C	6.7.29	A
6.7.30	B	6.7.31	C	6.7.32	C;D	6.7.33	B	6.7.34	A
6.7.35	B	6.7.36	D	6.7.37	A	6.7.38	D	6.7.39	A
6.7.40	A	6.7.41	A	6.7.42	A	6.7.43	X	6.8.1	5:5
6.9.1	C	6.9.2	N/A	6.9.3	N/A	6.9.4	N/A	6.9.5	A;C
6.9.6	B	6.9.7	B	6.9.8	B	6.9.9	A	6.9.10	D
6.9.11	B	6.9.12	B	6.9.13	C	6.9.14	D	6.9.15	C
6.9.16	D	6.9.17	C	6.9.18	D	6.9.19	C	6.9.20	B
6.9.21	D	6.9.22	B	6.9.23	C	6.9.24	A	6.9.25	B;C;D
6.9.26	B;C;D	6.9.27	A;B;C	6.9.28	A;C;D	6.9.29	D	6.9.30	A
6.9.31	B	6.10.1	C	6.11.1	0	6.11.2	N/A	6.11.3	N/A
6.11.4	B	6.11.5	N/A	6.11.6	B	6.11.7	B	6.11.8	D
6.11.9	B	6.11.10	D	6.11.11	A	6.11.12	B	6.11.13	C
6.11.14	B	6.11.15	A	6.11.16	1	6.11.17	3	6.11.18	B
6.11.19	2	6.11.20	4	6.11.21	8	6.11.22	D	6.11.23	120

6.11.24	4	6.11.25	A	6.12.1	A;C
6.12.4	D	6.12.5	B	6.12.6	C
6.14.1	D	6.14.2	D	6.15.1	N/A
6.15.4	A	6.15.5	N/A	6.15.6	N/A
6.15.9	D	6.15.10	50 : 50	6.15.11	A
6.15.14	B	6.15.15	C	6.16.1	A;D
6.16.4	D	6.16.5	A	6.16.6	B
6.16.9	B	6.16.10	C	6.16.11	D
6.16.14	C	6.16.15	D	6.16.16	A
6.18.1	N/A	6.18.2	A;C	6.18.3	A
6.18.6	C	6.18.7	D	6.18.8	C;D
6.18.11	C	6.18.12	A	6.18.13	C
6.18.16	3	6.18.17	B	6.18.18	X
6.18.21	C	6.18.22	C	6.18.23	44
6.18.26	B	6.18.27	A	6.18.28	C
6.19.2	C	6.19.3	A	6.20.1	True
6.20.4	D	6.20.5	C	6.20.6	D
6.20.9	X	6.20.10	A	6.20.11	D
6.20.14	A	6.20.15	C	6.20.16	C
6.20.19	A	6.20.20	A	6.20.21	A
6.20.24	B	6.20.25	6	6.20.26	D
6.20.29	D	6.20.30	C	6.20.31	A;C
6.20.34	A	6.20.35	D	6.21.1	D

6.12.2	C	6.12.3	B
6.13.1	6:6	6.13.2	B;D
6.15.2	A	6.15.3	N/A
6.15.7	D	6.15.8	B
6.15.12	B	6.15.13	D
6.16.2	A	6.16.3	D
6.16.7	D	6.16.8	D
6.16.12	B	6.16.13	D
6.17.1	C	6.17.2	C
6.18.4	N/A	6.18.5	C
6.18.9	D	6.18.10	N/A
6.18.14	B	6.18.15	B
6.18.19	A;B;C	6.18.20	D
6.18.24	15	6.18.25	C
6.18.29	C	6.19.1	N/A
6.20.2	False	6.20.3	B;C
6.20.7	B	6.20.8	N/A
6.20.12	C	6.20.13	B
6.20.17	A	6.20.18	C
6.20.22	C	6.20.23	2
6.20.27	C	6.20.28	C;D
6.20.32	C	6.20.33	B